

OPERATING MANUAL

HS-736USB

FT-736R CAT INTERFACE

Arduino / ATmega328P edition

HAM SPOT INC

<http://hamspot.com/>

FT-736R to computer CAT -- USB

NOTICE

This booklet is the operating manual for the **HS-736USB** computer-aided transceiver interface: a small USB box that lets modern amateur-radio software talk to a Yaesu **FT-736R** as if the radio were an **FT-847**. It also covers the few front- and rear-panel operations you need on the 736 itself so that CAT control will work.

It is **not** a Yaesu publication and does not replace the FT-736R Operating Manual or Technical Supplement. FT-736R, FT-847 and YAESU are trademarks of Yaesu Musen Co., Ltd. Photographs of the transceiver in this booklet are short excerpts from the original Yaesu FT-736R Operating Manual, reproduced here under fair use for identification and instruction only. Those images remain copyright Yaesu Musen.

The HS-736USB firmware (PIC original by N6BIL, Arduino port in this repository) is licensed under the GNU General Public License version 3. Hardware CAD, this booklet, and the remainder of the Ham Spot tree follow the same license unless a file says otherwise.

Credits: Dave KA6BFB (software proof of concept), Chuck N6BIL (PIC hardware translator, GPL), Ham Spot Inc (HS-736USB product), and the Arduino / ATmega328P port in **Arduino!**.

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Section 1. GENERAL DESCRIPTION



Fig. 1 Yaesu FT-736R VHF/UHF multimode base (excerpt from the Yaesu Operating Manual).

1.1 The HS-736USB Interface

The FT-736R remains a first-class satellite and VHF/UHF station. Its CAT (Computer Aided Transceiver) jack is TTL serial at **4800 bit/s, 8 data bits, no parity, 2 stop bits**. Many current programs — Ham Radio Deluxe in particular — never learned the 736 dialect. They *did* learn the later FT-847.

The HS-736USB sits between the computer and the radio. On USB it presents an FT-847 CAT port. On the DIN cable it speaks native FT-736R CAT, including the 23 cm mapping the 847 never had. Features the 736 does not have are ignored. Memory channels are **not** supported: put the radio on a VFO, and satellite mode on VFO, before you turn CAT on.

Two hardware generations exist. The original Ham Spot box is a PIC18F14K50 in a Hammond enclosure (see **Firmware/** and **Hardware/Eagle/**). The Arduino / ATmega328P port in **Arduino/** is a drop-in replacement of that translator. The **Nano carrier** in **Hardware/tscircuit/** is the board that generation sits on: the Nano plugs in; CAT, PTT MOSFET, RCA sense, dialect jumper, LCD, band and S-meter PWM are on the carrier. User operation of CAT is the same on PIC and Arduino.

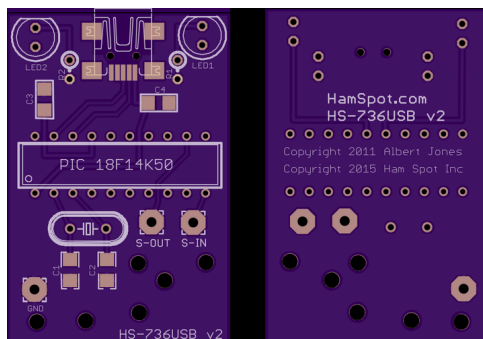


Fig. 2 Original HS-736USB printed circuit (v2 artwork, PIC18F14K50). USB Mini-B to the computer; S-IN / S-OUT / GND to the radio CAT jack.

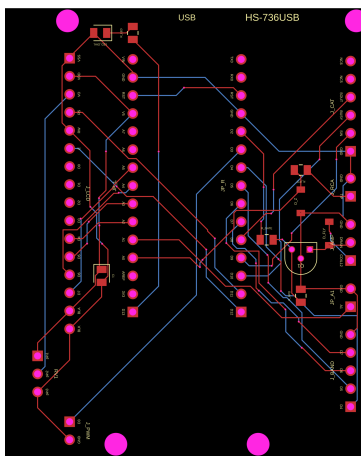


Fig. 2A Arduino Nano carrier (tscircuit). The Nano plugs into JP_L / JP_R (USB at the top). J_CAT is the 6-pin AMSAT CAT jack; pin 6 is not connected. Power is Nano USB 5 V. Red = top copper, blue = bottom.

1.2 What You Need

- An FT-736R with CAT enabled in the usual way (see §4).
- The HS-736USB (PIC cabinet or Arduino Nano carrier), powered from USB.
- A 6-pin DIN 240° (270° key) cable, wired to the AMSAT map in Appendix A — **not** blindly from a photocopied Yaesu pin-number drawing.
- Station software that can speak Yaesu FT-847 CAT at 4800 8N2 (Ham Radio Deluxe, Hamlib -m 1001, SatPC32, WSJT-X, ...), or native FT-736 CAT (-m 1010) if you put the Arduino in 736 dialect.

Section 2. THE FT-736R AT A GLANCE

This section is a short tour of the controls you will actually touch when the interface is in the station. It is not a substitute for Section 2 of the Yaesu Operating Manual.

2.1 Front Panel

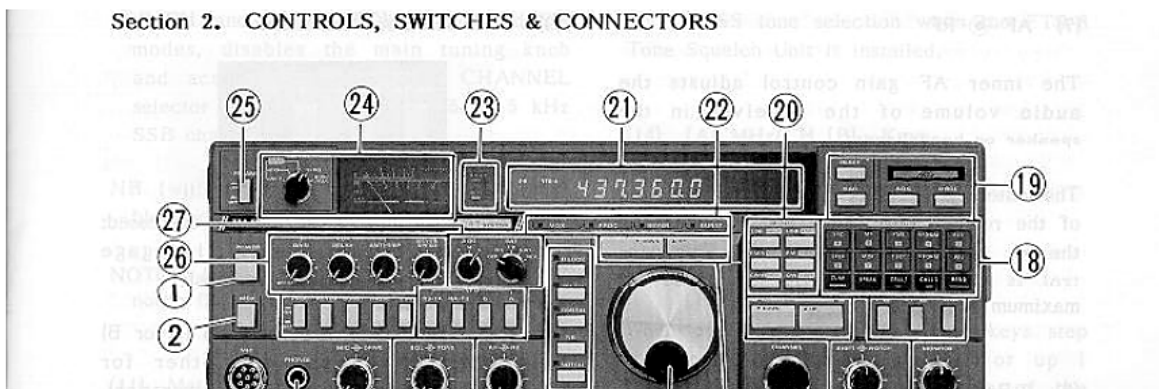


Fig. 3 Front panel, numbered as in the Yaesu Operating Manual (excerpt).

(1) POWER

Turns the transceiver on and off. Apply AC (or 13.8 V DC) and leave POWER on before you plug the interface into the computer; CAT ON from the PC will then lock the front panel.

(2) MOX

Manual transmit. With CAT running, Ham Radio Deluxe's on-screen PTT can drop the radio out of satellite mode — a long-standing HRD bug. The reliable workaround is to transmit with **MOX**, the microphone PTT, or a footswitch on the rear RCA, not with the program's PTT button. See §5.3.

(3) MIC jack

8-pin locking connector for the stock MH-1B8 (or MD-1B8 desk mic). Pin 2 is earth on the 736, not +5 V. An unmodified MH-31B8 does **not** drop in.

(11) Main Tuning Knob · (21) Display

Once CAT is on, the front panel is locked and the computer owns frequency and mode. Set the radio to a VFO (not a memory channel) **before** CAT ON. 1240 MHz is shown to FT-847 software as 240 MHz (the 847 has no 4 MHz digit).

(27) AGC and SAT Selectors



Fig. 4 AGC and SAT selectors (excerpt from the Yaesu Operating Manual).

(27) SAT selector

Full-duplex satellite operation. **OFF** is simplex. **RX / TX** display and tune one side while the other is fixed. **NOR** tracks both VFOs the same way; **REV** tracks them in opposite directions (the usual amateur-satellite inverting transponder). Leave this on a VFO-related SAT position, not a memory, before enabling CAT. The computer then turns full duplex on and off with the 847 **4E / 8E** (translated to 736 **0E / 8E**).

2.2 Rear Panel

2.3 REAR PANEL JACKS & SWITCHES

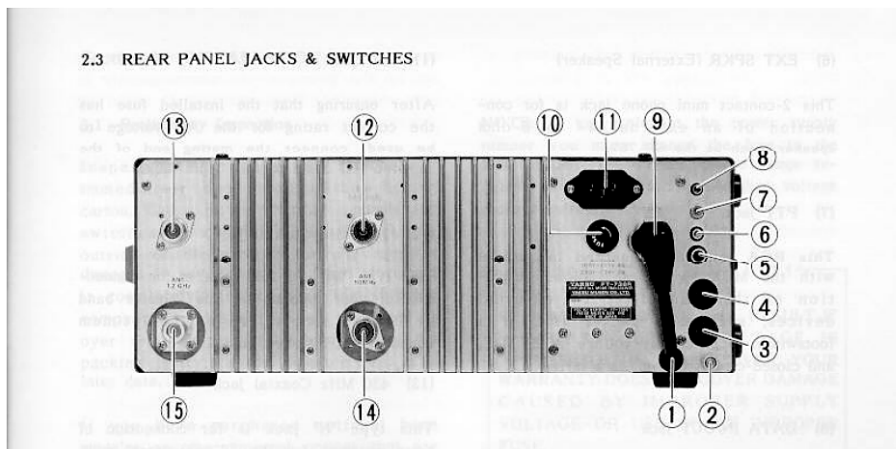


Fig. 5 Rear panel jacks (excerpt from the Yaesu Operating Manual). CAT is item (3), the 6-pin DIN above the KEY jack.

Typical layout, viewed from behind: DC pigtail from the internal switcher and the white 13.8 V inlet; ground screw; **CAT** 6-pin DIN 240°; **STBY** 5-pin DIN (amplifier ground-on-TX); **KEY**; **EXT SPKR**; **PTT RCA**; **DATA IN/OUT**; fuse; AC inlet; antenna jacks (N on 430, SO-239 on 144; optional modules below).

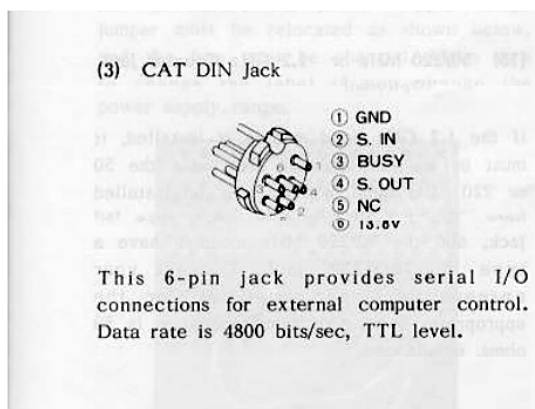


Fig. 6 CAT DIN as printed in the Yaesu Operating Manual (4800 bit/s, TTL). Use the AMSAT pin map in Appendix A — some printings number the shell differently.

MANUAL ERRORS

Do not trust a random photocopy of the operating manual for CAT or DATA pin numbers. Some printings number DIN pins with a Japanese / Yaesu order instead of IEC 60130, and some print the DATA jack tip/ring backwards. Identify CAT pins by the keyway plus the AMSAT map (Appendix A). On the radio, S.IN is data **into** the 736 (adapter TX); S.OUT is data **out of** the 736 (adapter RX).

Section 3. INSTALLATION OF THE INTERFACE

3.1 Powering the Radio



Fig. 7 AC inlet, DC pigtail and 13.8 V socket (excerpt from the Yaesu Operating Manual).

Before connecting mains, read the voltage label on the rear panel (100 / 117 / 220 / 234 V versions exist) and confirm the fuse matches that range (4 A on the low range, 2 A on the high range). For AC operation plug the internal-supply pigtail into the white 13.8 V inlet, then the AC cord. An external 13.8 V supply must be able to deliver 8 A continuous.

- !!!!! CAUTION !!!!! -

PERMANENT DAMAGE WILL RESULT IF IMPROPER SUPPLY VOLTAGE IS APPLIED TO THE TRANSCEIVER. YOUR WARRANTY DOES NOT COVER DAMAGE CAUSED BY IMPROPER SUPPLY VOLTAGE OR USE OF AN IMPROPER FUSE. DO NOT CONNECT THE MAINS CORD UNTIL ALL OTHER INTERCONNECTIONS HAVE BEEN MADE.

3.2 Connecting the HS-736USB

Power the interface from USB only. Wire a 6-pin DIN plug as follows (rear of the **male** plug, pins toward you; numbers are **not** sequential around the shell):

DIN pin	Radio name	Interface
1	GND	GND
2	S.IN (serial data in)	TX to the radio (Arduino D9)
3	BUSY	PIC: no connect. Nano J_CAT: D10
4	S.OUT (serial data out)	RX from the radio (Arduino D8)
5	NC	no connect
6	+13.8 V	do not connect

PIC cabinet: the silkscreen already reads S-IN / S-OUT / GND; those pads go to pins 2, 4 and 1. Nano carrier: plug the 6-pin DIN into **J_CAT** (GND, SIN, BUSY, SOUT, NC, NC). Pin 6 is not connected. Arduino D9 is AltSoftSerial TX, D8 is RX (Timer 1). Levels are TTL idle-high, same as the original HS-736USB. Do not insert a MAX232 unless your particular radio only clocks inverted CAT.

- !!!!! CAUTION !!!!! -

DO NOT FEED CAT PIN 6 (+13.8 V) INTO THE ARDUINO 5 V PIN OR THE PIC VDD RAIL. POWER THE INTERFACE FROM USB. PIN 6 IS A RADIO SUPPLY OUTPUT, NOT A SIGNAL.

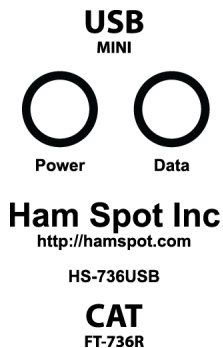


Fig. 8 Original HS-736USB cabinet legend (USB Mini-B, CAT to the FT-736R).

3.3 The Computer Port

Arduino / ATmega328P. The host USB-serial baud **must** be 4800 8N2 with no handshake. The original PIC CDC ignored the host baud; an Uno or Nano cannot. A genuine Uno (ATmega16U2) enumerates as Arduino USB serial; many Nano clones use a CH340. Install the matching driver if the operating system does not already have one. No Microchip **mchpcdc.inf** is required.

Original PIC HS-736USB. Windows enumerates the box as “CDC RS-232 Emulation Demo”. The 2014 booklet walked through the Found New Hardware Wizard and Microchip’s INF. That procedure applies only to the PIC cabinet, not to the Arduino port.

NOTE

Host settings in Hamlib, HRD, SatPC32 and WSJT-X: radio model **Yaesu FT-847**, **4800** baud, 8 data bits, no parity, 2 stop bits, no handshake. Default Arduino dialect is FT-847 (pin A1 open / HIGH). Ground A1, or send the CAT override in \$6, for native FT-736.

Section 4. BASIC OPERATION OF THE RADIO WITH CAT

4.1 Before You Enable CAT

1. POWER the radio. Confirm receive audio and that the correct antenna jack is in use for the band you intend (144 SO-239, 430 type N, optional modules on the lower pair).
2. Select a **VFO** with the VFO / MR keys. Do not leave the radio on a memory channel. The interface does not handle memories.
3. If you will run satellite, set the SAT selector to a VFO position (RX, TX, NOR or REV) — not a duplex memory. Simplex: SAT to OFF.
4. Plug in the CAT cable and the USB cable. The activity LED (Arduino D13, or the PIC cabinet LED) will blink when bytes move.

4.2 CAT ON and the Front Panel Lock

When the computer sends CAT ON (**00**), the 736 locks its front panel. In FT-847 dialect the adapter also forces 145.000 MHz FM onto the radio as the 847 CAT-ON extra; your logging program then writes the real frequency. CAT OFF (**80**) unlocks the panel. Do not fight the lock with the D LOCK button — turn CAT off from the PC if you need the knobs.

4.3 Simplex

Set band and mode from the program (or, before CAT ON, from the MODE buttons: LSB, USB, FM-N, FM, CW-N, CW). Tune from the program. PTT is CAT **08** on / **88** off, or MOX / mic / rear RCA. Repeater shift and CTCSS are translated; DCS from 847-style commands is dropped. The 736 simplex opcode is **89** — **88** is PTT off, despite some old 736 charts.

4.4 Satellite (full duplex)

With the adapter in FT-847 dialect (the usual HRD / SatPC32 / Hamlib 1001 case), the PC sends 847 satellite opcodes; the box rewrites them to 736:

What you want	PC (FT-847)	Radio (FT-736R)
Full duplex on	4E	0E
Sat RX frequency	11	1E
Sat TX frequency	21	2E
Sat RX / TX mode	17 / 27	same opcodes
Full duplex off	8E	8E then restore main VFO
Tone	0B	FA

If the PC is speaking native 736 while the adapter is in 847 mode, those 736 sat opcodes are **dropped with no error**. Simplex will still appear to work. That is the usual “SatPC32 won’t go duplex” failure — the program is set to FT-736 instead of FT-847. See §6.

4.5 23 cm

The FT-847 has no 4 MHz digit, so 1240–1299.99999 MHz is shown to the PC as 240–299.99999 MHz. The adapter adds **0xA0** to the first frequency BCD byte when it writes the radio. Tune 23 cm in the program as 240 MHz, not 1240 MHz.

4.6 After the QSO

Send CAT OFF before you unplug USB or power the radio down. If the panel stays locked, cycle POWER on the 736; that clears CAT as surely as **80**.

Section 5. COMPUTER SOFTWARE

5.1 Ham Radio Deluxe

On the Connect window, open the **Serial Ports** tab and note which COM (or /dev/ttyUSBn, /dev/ttyACM0) the interface received. Create a new connection: Company **Yaesu**, Radio **FT-847**, that port, speed **4800**, CTS/DTR/RTS off.

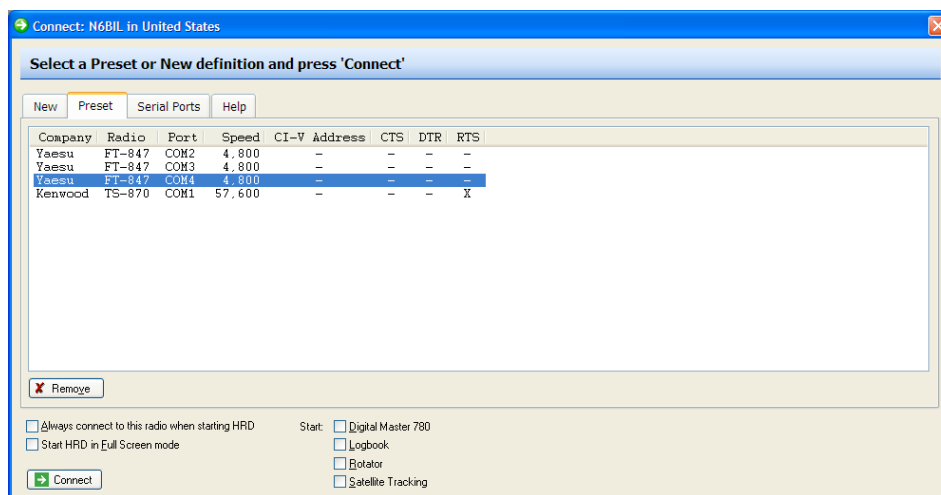


Fig. 9 Ham Radio Deluxe Connect window — Yaesu FT-847 at 4,800 (from the original HS-736USB booklet).

5.2 Hamlib, SatPC32, WSJT-X

Hamlib: **rigctl -m 1001 -r PORT -s 4800** for 847 dialect (default). Native 736 is **-m 1010** and requires the adapter in 736 dialect (\$6). SatPC32 and WSJT-X: pick FT-847, 4800 8N2, no flow control. Do not raise the radio CAT baud — the 736 stays at 4800; the adapter does not change it.

5.3 Known Issues with HRD

SAT PTT. If you press SAT on the radio, HRD will show SAT. If you then use HRD's software PTT, HRD may drop out of SAT mode while the 736 is still in SAT. Workaround: transmit with the radio MOX button, the mic PTT, or a footswitch. Do not use the on-screen PTT while in satellite mode.

Two-band satellite swap. Changing TX/RX bands in SAT via HRD is awkward on a two-band 736. A three- or four-band radio is easier. Example: TX on 2 m and RX on 70 cm — park TX on 6 m / 220 / 1200 briefly, move RX to 2 m, then put TX on 70 cm.

Section 6. FT-847 AND FT-736R DIALECTS

The Arduino port (not the original PIC) can speak either language on USB. The radio is always driven as a 736. Pin **A1** is an input with pull-up:

A1	USB dialect	Use with
HIGH (open)	FT-847 (default)	Hamlib 1001, HRD, SatPC32 as FT-847
LOW (jumper to GND)	FT-736R native	Hamlib 1010, programs that speak 736

A pin-change interrupt switches immediately and drops any half-finished 5-byte USB frame so the two dialects are never mixed. Frequency, PTT and sat flags in RAM stay put — you may flip the jumper mid-QSO without sending CAT OFF. Bounce is debounced 50 ms.

6.1 CAT Override

A non-Yaesu 5-byte block can ignore A1 for a while. It is accepted in both dialects:

A5 73 36 <sel> FC

<sel>	Source	Effect
00	Jumper	Follow A1 again
01	Force 847	847 even if A1 is low

<sel>	Source	Effect
02	Force 736	736 even if A1 is high
FF	Cycle	jumper -> 847 -> 736 -> jumper

Two stable jumper flips inside one second (wiggle the pin) cancel a CAT force and return to jumper-follow. Nothing in this block is sent to the radio.

6.2 If You Speak the Wrong Dialect

The adapter will not NAK. Unknown opcodes are dropped with no USB error. Overlap is large enough that simplex can look fine while satellite is silently dead.

736 software into an 847-mode adapter: duplex ON **0E**, sat frequencies **1E/2E** and tone **FA** never reach the radio. 736 CTCSS-on **{00,00,00,00,0A}** is misread as CAT ON and re-locks the panel.

847 software into a 736-mode adapter: **4E**, **11**, **21**, **0B** are locked out. HRD/SatPC32 satellite will not work until A1 is high again. Native 736 sat *does* work in this mode, and **E7/F7** are queried from the radio instead of a dummy 847 cache.

Section 7. ACCESSORY GPIO (ARDUINO / NANO CARRIER)

These pins are extras on the Uno/Nano port. They are not FT-847 CAT. On the Nano carrier they are headers next to the socket (Fig. 2A / Fig. 10). Prefer the radio's own STBY 5-pin DIN for band-keyed linears if you can.

Pin	Dir.	Function
D2	out	PTT MOSFET gate. HIGH when CAT PTT or RCA sense is keyed
D3	PWM	S-meter (0–255), Timer 2
D4–D7	out	One-hot 50 / 144 / 220 / 430 MHz
D10	in	CAT BUSY (LOW = squelch open)
D11	in	MCP23017 INTA (encoder); unused if expander off
A4, A5	I2C	SDA/SCL to MCP23017 (bare HD44780 + encoder)
A0	in	Optional PTT sense (active LOW)
A1	in	Dialect: HIGH = 847, LOW = 736
D8 / D9	UART	CAT RX / TX (AltSoftSerial)
D13	out	Dialect LED (HIGH = 847)

No 2-bit band outputs. 1240 MHz is shown on the LCD (host CAT as 240 MHz). In satellite, one-hot drives **both** RX and TX among 50/144/220/430.

Carrier jack	Pins	Goes to
J_CAT	GND SIN BUSY SOUT NC NC	FT-736R CAT DIN (pin 6 open)
J_AMP	COIL12 DRAIN GND	amp PTT MOSFET drain / coil +12
J_RCA	TIP GND	radio PTT RCA sense (22 k + zener)
JP_A1	A1 GND	open = 847 dialect; jumper = 736
J_BAND	D4 D5 D6 D7 GND	one-hot 50 / 144 / 220 / 430
J_PWM	D3 GND	S-meter PWM

Carrier jack	Pins	Goes to
J_LCD	16-pin HD44780	parallel from MCP23017; RV1 contrast; no backpack
J_ENC	5 V GND A B SW	KY-040; short press digit, long press mode

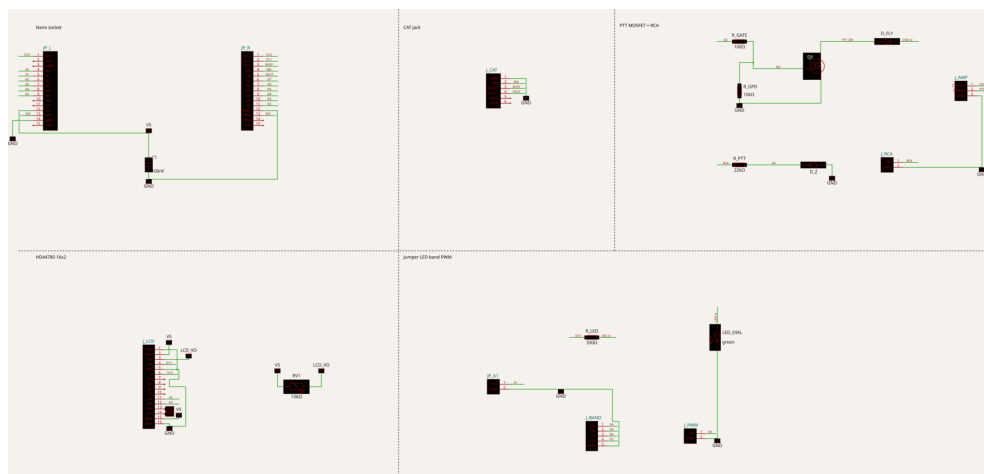


Fig. 10 Nano carrier schematic (same netlist as Fig. 2A). Nano socket, CAT jack, PTT MOSFET + RCA, HD44780, dialect jumper / LED / band / PWM.

7.1 PTT MOSFET

D2 -- 100 ohm -- MOSFET gate, 10 kohm to ground. Source to Arduino GND (common with radio GND). Drain to the amplifier PTT / relay (other side of the coil to +12 V, diode across the coil). A 2N7000 is enough for a small relay; use a logic-level power MOSFET for bigger coils. Output HIGH = TX = ground the amp PTT line, same sense as the 736 STBY jack. CAT **08/88** or the sense input keys this pin, so mic / MOX / footswitch keys the amp even when CAT PTT is idle.

7.2 Rear RCA PTT Sense (optional)

The 736 PTT RCA is 8 V open, ground-to-TX, 8 mA closed. It is in parallel with MOX and the mic PTT. **8 V will destroy a 328 pin.** Y-cable the RCA: one leg still to a footswitch if you use one; the other through 22 kohm to A0, with a 5.1 V zener to GND (cathode at A0). Firmware: INPUT_PULLUP, TX = pin LOW. Unplugged = RX.

- !!!!! CAUTION !!!!! -

DO NOT WIRE THE 8 V PTT RCA DIRECTLY TO A0. USE THE 22 kohm SERIES RESISTOR AND 5.1 V ZENER. CAT PTT STILL KEYS THE RADIO OVER SERIAL; THE MOSFET FOLLOWS CAT PTT OR THIS SENSE LINE.

7.3 S-Meter PWM

The adapter queries the 736 with CAT **F7** (the real radio command, not the dummy 847 **F7** on USB). Polling is every 3 s when idle, and about 633 ms (three times the CAT send time) when you have keyed in the last five minutes or the meter is above about S3. D3 PWM slews from the last sample to the new one across that window so an analog meter does not jump. Host CAT still wins the UART if a command arrives mid-poll.

Section 8. IN CASE OF DIFFICULTY

Symptom	Likely cause	What to do
No COM port	Driver, cable, or board	PIC: Microchip CDC INF. Arduino: 16U2 or CH340 driver. Try another USB cable that actually has data wires.

Symptom	Likely cause	What to do
Port exists, radio ignores CAT	Baud, DIN map, or CAT OFF	Host must be 4800 8N2 on Arduino. Confirm pin 2 = adapter TX, pin 4 = adapter RX, pin 1 = GND. Send CAT ON. Do not use pin 6.
Simplex works, sat does not	Wrong dialect	A1 HIGH and software = FT-847. Native 736 opcodes are dropped in 847 mode.
Panel locked, knobs dead	CAT is on	CAT OFF from the PC, or POWER-cycle the 736.
HRD drops SAT on PTT	HRD bug	Use MOX, mic PTT, or footswitch. Do not use HRD's PTT in SAT.
23 cm is 1 GHz off	847 digit limit	Tune 240 MHz in the program for 1240 MHz on the air.
Arduino dies when RCA is plugged in	8 V on A0	Add the 22 kohm + zener. Do not continue on a damaged 328.
S-meter PWM stuck	Poll lost the UART	Host CAT has priority. Idle the program; the next F7 should move D3.

Appendix A. CAT PIN MAP (USE THIS)

Wire from the drawing below (Fig. 6, repeated) and this table. AMSAT W6SHP: some FT-736R manuals and an old Yaesu application note numbered the CAT shell sequentially; the pins are not. Identify the keyway, then the numbered pins on the plug drawing.

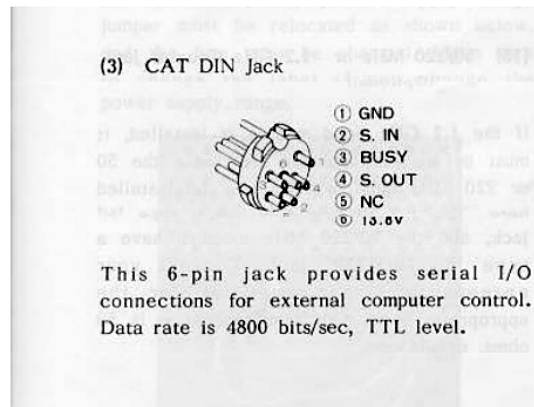


Fig. 6 (repeated). CAT 6-pin DIN from the Yaesu Operating Manual. Pin functions 1–6 match AMSAT; do not recount around the shell.

Pin	Radio name	HS-736USB / Arduino
1	GND	GND
2	S.IN (serial data in)	TX to the radio (D9)
3	BUSY	PIC: no connect. Nano: D10 (J_CAT)
4	S.OUT (serial data out)	RX from the radio (D8)
5	NC	no connect
6	+13.8 V	do not connect

Looking into the radio's CAT socket (holes), the pattern is mirrored left-right. DATA IN/OUT on the 3.5 mm jack is a separate story: tip = received data from the radio, ring = transmit data to the radio, sleeve = GND. Some manuals swap tip and ring in the drawing. Full front- and rear-panel pinouts: [Arduino/docs/CONNECTIONS.md](#).

Appendix B. SPECIFICATIONS OF THE INTERFACE

Item	PIC HS-736USB	Arduino Nano carrier
Host CAT	FT-847, 4800 8N2 (CDC ignores baud)	FT-847 or FT-736, baud must be 4800 8N2
Radio CAT	736, 4800 8N2 TTL, 50 ms inter-byte	same, on J_CAT (pin 6 open)
23 cm to PC	shown as 240 MHz	same
Memories	not supported; VFO only	same
Dialect switch	847 only	JP_A1 (A1–GND), IRQ, CAT A5 73 36 sel FC
PTT accessory	—	J_AMP MOSFET; optional J_RCA sense
Band decode	—	J_BAND one-hot D4–D7 (50/144/220/430)
S-meter	—	J_PWM D3, slewed, radio F7 poll
Display	—	J_LCD 16×2 HD44780, RV1 contrast
MCU board	PIC18F14K50 on the PCB	Arduino Nano in JP_L / JP_R
CAD	Hardware/Eagle	Hardware/tscircuit
Firmware	Firmware/Source (GPL-3)	Arduino/firmware (GPL-3)

Opcode notes: **Arduino/docs/**. Original PIC booklet: **Manual/HS-736USB-Manual.pdf**. Figs. 1 and 3–7: Yaesu FT-736R Operating Manual (fair use). Fig. 9: HS-736USB booklet. Figs. 2 and 8: Ham Spot. Figs. 2A and 10: Nano carrier (Hardware/tscircuit).